

Errata: Practical Uncertainty in Neural Networks

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p37, Eqn 3.30 for leap frog integration, the step should be $\epsilon/2$. I am not happy with the notation there. Instead use,

$$\begin{aligned}v_i(t + \frac{\epsilon}{2}) &= v_i(t) + \frac{\epsilon}{2} \frac{dv_i(t)}{dt}, \\w_i(t + \epsilon) &= w_i(t) + \epsilon \frac{dw_i(t + \epsilon/2)}{dt}, \\v_i(t + \epsilon) &= v_i(t + \frac{\epsilon}{2}) + \frac{\epsilon}{2} \frac{dv_i(t + \epsilon)}{dt}.\end{aligned}$$

p51 For the α divergence on page 75, the limits to relate to forward/reverse KL divergence and Hellinger distance were not correct. It should be, $KL(q|p) : \alpha \rightarrow 0$, $KL(p|q) : \alpha \rightarrow 1$, Hellinger: $\alpha \rightarrow 0.5$.

p51 Eq. 3.58 missing factor of $1/2$

p107, Eq 4.7: $p(\mathbf{w}|\mathbf{X}, \mathbf{Y}, \theta) \propto \left[\prod_{i=0}^{N-1} \mathcal{N}(\mathbf{y}_i | f(\mathbf{x}_i; \theta) \mathbf{w}, \sigma^2) \right] p(\mathbf{w})$.

Eq 4.11 and 4.15: are missing Φ , they should read

$\sigma_N^2(\hat{\mathbf{x}}) = \sigma^2 + \Phi(\hat{\mathbf{x}})^T \Sigma_\pi \Phi(\hat{\mathbf{x}})$ and $\sigma_N^2(\hat{\mathbf{x}}) \approx \sigma^2 \Phi(\hat{\mathbf{x}})^T \Sigma_{\text{diag}} \Phi(\hat{\mathbf{x}})$

p111, Eq 4.22: The approximation from ref [34] in the paper provides the Taylor Series approximation of $\mathcal{N}(\frac{\mu_j}{\mu_d}, \frac{\mu_j^2}{\mu_d^2} (\sigma_j^2/\mu_j^2 + \sigma_d^2/\mu_d^2))$. The issue with this is the square of the μ_d terms, as this is at risk of numerical overflow in half floating-point precision. Instead we use in $\sigma^2 \approx \mu_j/\mu_d \sqrt{\sigma_j^2 + \sigma_d^2}$, which empirically followed a similar trend and was less of a risk for numerical overflow (though not immune).

p135, Eq. 5.13, the integral sign should be there in the denominator.

p141, Eq. 5.34, replace η with $\eta/2$

p143, Eqn 5.43, and **Pg. 146** Eqn 5.46, Typo in missing square term (highlighted in red).

$$\begin{aligned}& \frac{\sigma^2}{2} \int_{-\infty}^{\infty} v^2 \operatorname{erf}\left(\frac{(\sigma v + \mu)}{\sqrt{2}}\right) \phi(v) dv \\&= \frac{\sigma^2}{4\sqrt{2\pi}} \left[\frac{\sqrt{\pi}}{\alpha_1 \sqrt{\alpha_1}} \operatorname{erf}\left(\frac{\sqrt{\alpha_1} b_1}{\sqrt{\alpha_1 + a_1^2}}\right) - \frac{1}{\alpha_1 \sqrt{\alpha_1 + a_1^2}} \frac{2a_1^2 b_1}{\alpha_1 + a_1^2} \exp\left(\frac{-\alpha_1 b_1^2}{\alpha_1 + a_1^2}\right) \right]\end{aligned}$$

This same typo carried over to integral property from [240, EQ. 2.7.3.9]

$$\int_{-\infty}^{\infty} z^2 \exp(\alpha^2) \operatorname{erf}(\alpha_1 z + \beta_1) = \frac{\sqrt{\pi}}{2\alpha\sqrt{\alpha}} \operatorname{erf}\left(\frac{\sqrt{\alpha}\beta_1}{\sqrt{\alpha + \alpha_1^2}}\right) - \frac{1}{\sqrt{\alpha + \alpha_1^2}} \frac{2\alpha_1^2\beta_1}{\alpha + \alpha_1^2} \exp\left(\frac{-\alpha\beta_1^2}{\alpha + \alpha^2}\right)$$

And to the final result on **Pg. 147**, Eqn 5.61 for the approximation to the variance of the Gelu function,

$$\begin{aligned} \sigma_g^2 \approx & \frac{\sigma^2 + \mu^2}{2} \\ & + \frac{\sigma^2}{4\sqrt{2\pi}} \left[\frac{\sqrt{\pi}}{\alpha_1\sqrt{\alpha_1}} \operatorname{erf}\left(\frac{\sqrt{\alpha_1}b_1}{\sqrt{\alpha_1 + a_1^2}}\right) - \frac{1}{\alpha_1\sqrt{\alpha_1 + a_1^2}} \frac{2a_1^2b_1}{\alpha_1 + a_1^2} \exp\left(\frac{-\alpha_1b_1^2}{\alpha_1 + a_1^2}\right) \right] \\ & + \frac{\mu\sigma a_2}{\alpha_2\sqrt{2\pi(a_2^2 + \alpha_2)}} \exp\left(-\frac{\alpha_2b_2^2}{a_2^2 + \alpha_2}\right) \\ & + \frac{\mu^2}{2\sqrt{2\alpha_3}} \operatorname{erf}\left(\frac{2\alpha_3b_3}{2\sqrt{\alpha_3^2 + \alpha_3a_3^2}}\right) \\ & - \frac{(2\alpha_4^2 + b_4^2)}{16\sigma\alpha_4^5\sqrt{2}} \exp\left(\frac{b_4^2}{4\alpha_4^2} + \gamma\right) - \mu_g^2 \end{aligned}$$

where, $\alpha_1, \alpha_2, \alpha_3 = \frac{1}{2}$, $b_1, b_2, b_3 = \frac{\mu}{\sqrt{2}}$, $a_1, a_2, a_3 = \frac{\sigma}{\sqrt{2}}$, $\alpha_4 = \sqrt{\frac{1}{2\sigma^2} + \frac{\eta}{2}}$, $b_4 = \frac{\mu}{\sigma^2}$ and $\gamma = -\frac{\mu^2}{2\sigma^2}$, $\eta = 1.23907$.

pg. 150, Eqn 5.68, the last term are wrong way around. Should read,

$$\Omega = \Omega_X \Omega_Y + \Omega_X \mu_Y^2 + \mu_X^2 \Omega_Y.$$